

Project Abstract — UniGPT

An AI Academic Copilot for Smart University Management

1. Problem Statement

Modern universities still rely heavily on fragmented systems and manual workflows for managing academic activities. Students frequently struggle to access important course materials, notices, assignment guidelines, attendance records, and academic policies because these resources are scattered across multiple disconnected platforms such as ERP systems, WhatsApp or Messenger groups, Google Classroom, spreadsheets, and paper-based records. Faculty members face similar challenges, spending significant time on repetitive tasks such as attendance tracking, assessment creation, grading, feedback generation, and academic communication. Administrative departments also lack real-time institutional visibility, making decision-making slow and inefficient.

2. Innovation & Uniqueness

UniGPT addresses this problem by introducing an intelligent, centralized academic platform powered by Artificial Intelligence. Rather than forcing students, faculty, and administrators to navigate multiple disconnected tools, UniGPT unifies all major academic workflows into a single platform with role-based experiences. The system provides one integrated ecosystem for academic communication, course management, attendance, assessments, document sharing, analytics, and AI-assisted decision support.

The key innovation of UniGPT lies in its Retrieval-Augmented Generation (RAG) architecture. Unlike generic AI chatbots such as ChatGPT that may generate inaccurate or hallucinated responses, UniGPT grounds every AI response in institution-approved documents such as syllabi, policies, handbooks, lecture materials, and academic resources. When a user asks a question, the system retrieves relevant document passages, performs contextual reasoning, and generates a precise answer with source citations and confidence scores. This ensures that responses are trustworthy, auditable, and aligned with institutional rules.

3. AI Usage

3a. AI Within the Product

UniGPT uses advanced Artificial Intelligence through Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to deliver accurate academic assistance. The platform integrates OpenAI models, specifically GPT-4o for reasoning and response generation, and text-embedding-3-large for semantic document search.

When a user asks a question, UniGPT first converts both the query and approved academic documents into vector embeddings. It then performs semantic similarity

search to retrieve the most relevant document chunks (such as syllabus pages, policies, lecture materials, or handbooks). These retrieved passages are provided as context to the language model, allowing the AI to generate responses grounded in institutional data rather than generic internet knowledge.

AI is used in multiple areas of the platform, including:

- Intelligent academic Q&A with source citations
- Automated quiz and assignment generation
- AI-assisted grading feedback
- Personalized student guidance
- Predictive academic analytics for risk detection

3b. AI-Assisted Development Tools

The development of UniGPT itself was powered by AI-assisted tools throughout the entire software development lifecycle. The following tools were used responsibly during development:

- Claude Code — Used as the primary AI coding assistant for backend (Laravel) and frontend (Vue.js/Inertia.js) implementation, code review, refactoring, and architecture decisions.
- ChatGPT — Used for initial idea generation, feature brainstorming, and drafting content structures.
- Gemini — Used for research on RAG architecture best practices, academic platform benchmarking, and exploring AI-in-education literature.

All AI-generated code was reviewed, understood, and validated by the development team. No AI output was blindly accepted or submitted without comprehension.

3c. Effective & Ethical Use of AI

UniGPT is committed to responsible and ethical AI use at every layer — both within the platform and during its development. The following principles guide our approach:

- **Permission-Aware Retrieval:** Users can only access documents they are authorized to view. The RAG system enforces role-based document boundaries, ensuring student, faculty, and admin data remain appropriately scoped.
- **Source Citations & Confidence Scores:** Every AI-generated response includes source citations and confidence indicators, reducing hallucination risk and enabling users to verify answers against original institutional documents.
- **No Hallucination Policy:** UniGPT does not rely on generic internet knowledge for academic answers. All responses are grounded in institution-approved content, making the system auditable and trustworthy.
- **Transparency in Development:** All AI tools used during development (Claude Code, ChatGPT, Gemini) are legal, publicly available, and policy-compliant.

The development team can fully explain and defend every AI-assisted decision.

- **Data Privacy:** No sensitive student or institutional data is used to train or fine-tune any external AI model. The system processes data in-context only, with retrieval limited to indexed, approved documents.
- **Human Oversight:** All AI outputs — whether in the product or during development — are reviewed by human team members before deployment.

4. Core Features

UniGPT offers comprehensive core features tailored to three major user groups. Students receive an AI academic copilot that helps them access course materials, track attendance, monitor GPA/CGPA, manage deadlines, and receive instant academic assistance. Faculty members gain AI-powered teaching tools for generating assessments, managing attendance, publishing materials, grading submissions, and analyzing course performance. Administrators gain centralized control over users, departments, academic structures, document approvals, permissions, announcements, and institution-wide analytics.

5. Feasibility

From a technical perspective, UniGPT is highly feasible and production-ready. The system is built using modern scalable technologies including Laravel 11, Vue.js 3, Inertia.js, Tailwind CSS, and MySQL. Its modular architecture supports secure role-based access control, efficient document indexing, AI provider abstraction, and maintainable long-term development. The system has already implemented major admin, faculty, and student functionalities, demonstrating strong real-world applicability.

6. Scalability & Future Vision

UniGPT is designed with scalability in mind. Its architecture can support thousands of students, faculty members, departments, and documents across multiple institutions. Future expansion includes real-time messaging, integration with messaging platforms like WhatsApp and Telegram, predictive analytics, voice-based AI interaction, digital library integration, and support for multiple AI providers. The long-term vision is to transform universities into AI-driven intelligent ecosystems where academic knowledge becomes instantly accessible, administrative friction is minimized, and educational productivity is significantly enhanced.